SECTION 23 07 13

DUCTWORK INSULATION

Rev #	Description of Change	Author	Package	Date
23 07 13				
A	Initial Issue; Issue for Final Review	Ankush Shelar	FP-03-01A-01	27 Jun 2025

PART 1 GENERAL

1.1 SUMMARY

A. Section includes requirements necessary to furnish and install ductwork thermal insulation and internal sound attenuation insulation.

1. Related Sections: Section 23 05 93 – Air Systems Testing, Adjusting and Balancing.
2. Section 23 07 13.01 – Ductwork Insulation, Schedule.
3. Section 23 31 01.01 – HVAC Ductwork Schedule.
4. Section 40 05 01 – Basic Mechanical Requirements.

B. CAUTION: Use of this Section without including the above-listed items results in omission of basic requirements.

C. In the event of conflict regarding Ductwork Insulation requirements between this Section and another section, the provisions of this Section govern.

1.2 REFERENCES

A. ASTM International (ASTM) (www.astm.org):
1. ASTM C411 – Standard Test Method for Hot-Surface Performance of High-Temperature Thermal Insulation.
2. ASTM C423 – Standard Test Method for Sound Absorption and Sound Absorption Coefficients by the Reverberation Room Method.
3. ASTM C518 – Standard Test Method for Steady-State Thermal Transmission Properties by Means of the Heat Flow Meter Apparatus.
4. ASTM C534 / C534M – Standard Specification for Preformed Flexible Elastomeric Cellular Thermal Insulation in Sheet and Tubular Form.
5. ASTM C871 – Standard for Chemical Analysis of Thermal Insulation.
6. ASTM C975 – Standard Practice for Preparing Test Specimens from Basic Refractory Ramming Products by pressing.
7. ASTM C1071 – Standard Specification for Fibrous Glass Duct Lining Insulation (Thermal and Sound Absorbing Material).

8. ASTM C1335 - Standard Test Method for Measuring Non-Fibrous Content of Man-Made Rock and Slag Mineral Fiber Insulation.
9. ASTM C1338 – Standard Test Method for Determining Fungi Resistance of Insulation Materials and Facings.
10. ASTM C1534 – Standard Specification for Flexible Polymeric Foam Sheet Insulation Used as a Thermal and Sound Absorbing Liner for Duct Systems.
11. ASTM D1667 – Standard Specification for Flexible Cellular Materials—Poly (Vinyl Chloride) Foam (Closed-Cell).
12. ASTM D330 / D3330M – Standard Test Method for Peel Adhesion of Pressure-Sensitive Tape.
13. ASTM D3611 – Standard Practice for Accelerated Aging of Pressure-Sensitive Tapes.
14. ASTM D3654 / D 3654M – Standard Test Methods for Shear Adhesion of Pressure-Sensitive Tapes.
15. ASTM D3816 / D 3816M – Standard Test Method for Water Penetration Rate of Pressure-Sensitive Tapes.
16. ASTM E84 – Standard Test Method for Surface Burning Characteristics of Building Materials.
17. ASTM E96 – Standard Test Methods for Water Vapor Transmission of Materials.
18. ASTM G7 / G7M – Standard Practice for Atmospheric Environmental Exposure Testing of Nonmetallic Materials.
19. ASTM G21 – Standard Practice for Determining Resistance of Synthetic Polymeric Materials to Fungi.

B. National Fire Protection Association (NFPA) (www.nfpa.org):
   1. NFPA 90A – Standard for the Installation of Air-Conditioning and Ventilating Systems.
   2. NFPA 90B – Standard for the Installation of Warm Air Heating and Air Conditioning Systems.
   3. NFPA 255 - Standard Method of Test of Surface Burning Characteristics of Building Materials.

C. UL (www.ul.com):
   1. CAN/ULC-S102 – Standard Method of Test for Surface Burning Characteristics of Building Materials and Assemblies.
   2. UL 181 – Standard for Factory-Made Air Ducts and Air Connectors.
   3. UL 723 – Standard for Test for Surface Burning Characteristics of Building Materials.
   4. UL 263 - Fire Tests of Building Construction and Material.

D. Manufacturing, Fabrication, and Installation Standards:
   1. SMACNA (4th Edition, 2020) – Sheet Metal & Air Conditioning Contractors National Association.

E. Energy Conservation Standards:
   1. ASHRAE 90.1 – Energy Standard for Buildings Except Low-Rise Residential Buildings.
   2. IECC-2003 – International Energy Conservation Code.

3. IMC – International Mechanical Code.
4. ECBC (India) – Energy Conservation Building code by Bureau of Energy Efficiency (BEE).

F. Examination Standard
   1. FM Approvals-Class number 4924

1.3 SUSTAINABLE DESIGN REQUIREMENTS

A. Comply with Sustainable Design Requirements, refer to Section 40 05 01 – Basic Mechanical Requirements.

1.4 DEFINITIONS

A. Clean Zone: Areas of the building defined by the column gridlines noted in Section 40 05 01 – Basic Mechanical Requirements.

1.5 SUBMITTALS

A. Refer to Section 40 05 01 – Basic Mechanical Requirements and the Submittal Schedule at the end of Part 3 for a list of submittal requirements.

1.6 QUALIFICATIONS

A. Manufacturer’s Qualifications: Manufacturer regularly engaged in the manufacturing of HVAC duct insulation of similar type to that specified for a minimum of 10 years.

B. Installer's Qualifications:
   1. Installer regularly engaged in installation of HVAC duct insulation of similar type to that specified for a minimum of 5 years.
   2. Use persons trained for installation of HVAC duct insulation in industrial applications.

C. Inspection & Verification:
   1. The use of certified mechanical insulation inspectors who maintain current certification by the National Insulation Association, or other certified mechanical insulation certification association, is recommended throughout the project to inspect and verify the materials are and the total insulation system has been installed in accordance with the specifications and material manufacturer’s instruction.

D. Fire-Test-Response Characteristics: Insulation and related materials must have fire-test-response characteristics indicated, as determined by testing identical products per ASTM E 84/UL723, by a testing and inspecting agency acceptable to authorities having jurisdiction. Factory label insulation and jacket materials and adhesive, mastic, tapes, and cement material containers, with appropriate markings of applicable testing and inspecting agency.
   1. Insulation Installed Indoors: Flame-spread index of 25 or less, and smoke-developed index of 50 or less.
   2. Insulation Installed Outdoors: Flame-spread index of 75 or less, and smoke-developed index of 150 or less.

E. Material with fire performance of Class 0 that conform to BS 476 Part 6 for Fire Propagation and class 1 as per BS 476 Part 7 for surface spread of flame.

F. Mineral wool/fiber material with fire performance of A1 Class Non-Combustible as per EN 13501 fire classification. Flame and smoke development index must be 0 as per UL723 / ASTM E84.

1.7 DELIVERY, STORAGE, AND HANDLING

A. Packaging: Insulation material containers must be marked by the manufacturer with appropriate ASTM standard designation, type and grade, and maximum use temperature.

B. Storage and Handling Requirements:
   1. Store and handle materials in accordance with manufacturer’s instructions.
   2. Keep materials in manufacturer’s original, unopened containers and packaging until installation.
   3. Store materials in clean, dry area indoors.
   4. Do not store materials directly on floor or ground.
   5. Store materials out of direct sunlight.
   6. Keep materials away from freezing conditions.
   7. Protect materials during storage, handling, and installation to prevent damage.

1.8 COORDINATION

A. Coordinate size and location of seismic supports, anchors, hangers, and insulation shields specified in Section 40 05 01 – Basic Mechanical Requirements.

B. Coordinate clearance requirements with duct installer for duct insulation application. Before preparing ductwork shop drawings, establish and maintain clearance requirements for installation of insulation and field-applied jackets and finishes and for space required for maintenance. Coordinate duct material and pressure requirements with Section 23 31 01.01 – HVAC Ductwork Schedule.

1.9 SCHEDULING

A. Schedule insulation application after pressure testing systems and, where required, after installing and testing heat tracing. Insulation application may begin on segments that have satisfactory test results. Refer to Section 23 05 93 – Air Systems Testing, Adjusting and Balancing for requirements.

B. Complete installation and concealment of plastic materials as rapidly as possible in each area of construction.

C. Do not install HVAC duct insulation under ambient conditions outside manufacturer's limits.

PART 2 PRODUCTS

2.1 GENERAL

A. Provide insulations conforming to ASHRAE 90.1, IECC and IMC.

B. Products must not contain asbestos, lead, mercury, or mercury compounds.

C. Products that come into contact with stainless steel must have a leachable chloride concentration below 50 ppm, when tested according to ASTM C871.

D. Insulation materials for use on austenitic stainless steel to be qualified as acceptable according to ASTM C795.

E. Foam insulation materials must not use CFC or HCFC blowing agents in the manufacturing process.

F. Comply with requirements in Section 23 07 13.01 Ductwork Insulation, Schedule where installing insulating materials.

G. Provide insulations conforming to FM 4924.

2.2 EXTERNAL DUCT INSULATION

A. Materials:
   1. Type A:
      a. Flexible glass-fiber insulation, thickness as scheduled, 12 kg per cubic meter density, with UL-labeled FSK foil vapor barrier.
      b. Thermal conductivity no greater than 0.041 watt per meter-kelvin at 24 degree C.
      c. Water vapor permeability (facing) not to exceed 0.02 Perm-in. or less per ASTM E96, procedure A, test method.
      d. Acceptable Manufacturers:
         1) CertainTeed Corp: Duct Wrap Insulation, Type 75.
         2) Johns Manville : R-Series Microlite.
         3) Knauf Fiber Glass: Knauf Duct Wrap.
         4) Owens Corning: Fiberglas All Service Duct Wrap.
   2. Type B:

a. Semi-rigid glass fiber insulation per ASTM C612 Standard Specification for Mineral Fiber Block and Board Thermal Insulation, thickness as scheduled, 48 kg per cubic meter with UL-labeled, foil-reinforced, kraft-faced vapor barrier12-micron thick aluminum foil with reinforced glass scrim.
b. Thermal conductivity no greater than 0.041 Watt per meter-kelvin at 24 degree C.
c. Water vapor permeability (facing) not to exceed 0.02 Perm-in. or less per ASTM E96, procedure A, test method.
d. Acceptable Manufacturers:
   1) CertainTeed Corp: Fiberglass Board Insulation, Type CB 300.
   2) Johns Manville: Type 814 Spin-Glas.
   3) Knauf Fiber Glass: Knauf Insulation Board.
   4) Owens Corning: Fiberglas Board Insulation, Type 703.

3. Type C:
   a. Rigid glass fiber insulation per ASTM C612, thickness / overall thermal resistance as scheduled, 96 kg per cubic meter density with UL-labeled, foil-reinforced, kraft-faced vapor barrier.
   b. Thermal conductivity no greater than 0.033 Watt per meter-kelvin at 24 degree C.
   c. Water vapor permeability (facing) not to exceed 0.02 Perm-in. or less per ASTM E96, procedure A, test method.
   d. Acceptable Manufacturers:
      1) CertainTeed Corp: Fiberglass Board Insulation, Type CB 600.
      2) Johns Manville: Type 817 Spin-Glas.
      3) Knauf Fiber Glass: Knauf Insulation Board.
      4) Owens Corning: Fiberglas Board Insulation, Type 705.
4. Type D:
   a. Semi-rigid, large diameter pipe, round duct and tank wrap shipped in roll form. The insulation density: 40 kg per cubic meter and complying with ASTM C1393, Type I, II, IIIA, IIIB, Category 2 and ASTM E84 FHC 25/50.
   b. Thermal conductivity no greater than 0.035 Watt per meter-kelvin at 24 degree C mean temperature.
   c. Water vapor permeability not to exceed 0.02 Perm-in. or less per ASTM E96, procedure A, test method.
   d. Acceptable Manufacturers:
      1) CertainTeed Corp: Fiberglass Board Insulation, Type CB 600.
      2) Johns Manville: Type 817 Spin-Glas.
      3) Knauf Fiber Glass: Knauf Insulation Board.
5. Type E:
   a. Indoor application: Flexible Elastomeric, FM 4924 approved fire inhibited Closed-cell Elastomeric Nitrile, Rubber materials with factory laminated with 12-micron thick aluminum foil with reinforced glass scrim. Comply with ASTM C534, Type I for tubular materials and Type II for sheet materials. Thickness determines type (NBR/PVC or EPDM) complying with 25/50 FSI/SDI limits when tested per ASTM E84 or UL723 standards.

b. Fire performance complies with Class ‘O’ as per BS 476 Part 6 for fire propagation and Class 1 as per BS 476 Part 7 for surface spread of flame.
c. Thermal conductivity is no greater than 0.035 Watt per meter-kelvin at 0 degree C mean temperature per ASTM C177 or C518 test methods.
d. Water vapor permeability for NBR/PVC/EPDM based material not to exceed 0.05 Perm-in. or less per ASTM E96, procedure A, test method.
e. Water vapor permeability for EPDM based material not to exceed 0.08 Perm-in. or less per ASTM E96, procedure A, test method.
f. CFC/HFC and Formaldehyde free material with density between 40 to 55 Kg per cubic meter and UV resistance as per ASTM G 26A / ISO4892-2 Method A.
g. Withstand maximum surface temperature of 85-degree C and minimum surface temperature of minus 50-degree C as per ASTM C534. Moisture Diffusion Resistance Factor or ‘μ’ value should be minimum 12,000 as per EN12086.
h. Density of insulation must be between 40 to 55 Kg per cubic meter.
i. Kg per cubic meter Acceptable Manufacturers:
    1) Armaflex
    2) K-Flex

6. Type F:
    a. Outdoor application: Flexible Elastomeric, FM 4924 approved fire inhibited Closed-cell Elastomeric Nitrile, Rubber materials with UV resistant metal finishing on outer or factory laminated with 12-micron thick aluminum foil with reinforced glass scrim. Comply with ASTM C534, Type I for tubular materials and Type II for sheet materials. Thickness determines type (NBR/PVC or EPDM) complying with 25/50 FSI/SDI limits when tested per ASTM E84 or UL723 standards.
    b. Fire performance complies with Class ‘O’ as per BS 476 Part 6 for fire propagation and Class 1 as per BS 476 Part 7 for surface spread of flame.
    c. Thermal conductivity is no greater than 0.035 Watt per meter-kelvin at 0 degree C mean temperature per ASTM C177 or C518 test methods.
    d. Water vapor permeability for NBR/PVC/EPDM based material not to exceed 0.05 Perm-in. or less per ASTM E96, procedure A, test method.
    e. Water vapor permeability for EPDM based material not to exceed 0.08 Perm-in. or less per ASTM E96, procedure A, test method.
    f. CFC/ HFC & Formaldehyde free material with density between 40 to 55 Kg per cubic meter and UV resistance as per ASTM G 26A / ISO4892-2 Method A.
    g. Withstand maximum surface temperature of 85-degree C and minimum surface temperature of 0-degree C as per ASTM C534.
    h. Moisture Diffusion Resistance Factor or ‘μ’ value should be minimum 60,000 as per EN12086.
    i. Density of insulation must be between 50 to 65 Kg per cubic meter.
    j. Water absorption by volume of insulation material less than 0.1%.
    k. Acceptable Manufacturers:
        1) Armaflex
        2) K-Flex

7. Type G
   a. Exterior application, FM approved flexible elastomeric thermal insulation with a white 17.5 mils laminated covering, with or without a pressure sensitive adhesive in sheet and roll form. Comply with ASTM C534, Type II – sheet grade 1 for foam only. Product density 48 – 96 kg per cubic meter per ASTM D1667.
   b. Thermal conductivity no greater than 0.035 Watt per meter-kelvin at 0 degree C, mean temperature per ASTM C177 or C518 test methods.
   c. Water vapor permeability not to exceed 0.05 Perm-in.(foam) or less per ASTM E96, procedure A, test method. Provide laminates that are impermeable.
   d. Water absorption to be no more than 0.2 percent by volume conforming to ASTM C1763, procedure B test methods.
   e. CFC/HFC & Formaldehyde free material with density between 40 to 55 Kg per cubic meter and UV resistance as per ASTM G 26A / ISO4892-2 Method A.
   f. Withstand maximum surface temperature of 85-degree C and minimum surface temperature of 0-degree C.
   g. UV/Mechanical Protection with UV Resistant Paint for Outdoor application (For ducting). Apply UV resistant paint as per manufacturer recommendation on insulation material. Spread 7 mil woven glass cloth/225 GSM chopped strand material (with minimum overlap of 2 inch at the joints) on the insulation while paint is wet. Apply another coat of the paint on the woven glass cloth & allowed it to dry.
   h. Acceptable Manufacturers:
      1) Armacell LLC; ArmaTuff.
      2) Approved equal in performance and quality

2.3 INTERNAL DUCT LINER

A. Materials:
   1. Type L1:
      a. Flexible duct liner insulation, thickness as scheduled, 24 kg per cubic meter density with fire-resistant fiberglass mat coating on the air side rated for maximum 6,000-fpm air velocity.
      b. Compliance with ASTM C1071 Type 1.
      c. Thermal conductivity no greater than 0.035 Watt per meter-kelvin at 24 degree C.
      d. Acceptable Manufacturers:
         1) CertainTeed Corp: Fiberglass Board Insulation, Type CB 600.
         2) Johns Manville: Type 817 Spin-Glas.
         3) Knauf Fiber Glass: Knauf Insulation Board.
         4) Owens Corning: Fiberglas Board Insulation, Type 705.

   2. Type L2:
      a. Flexible elastomeric duct liner: EPDM-rubber-based, fiber-free, microbial-resistant, acoustic, flexible, closed-cell. Complying with ASTM C1534, Type I and with NFPA 90A and NFPA 90B. Flame-

spread index of 25 and maximum smoke-developed index of 50 when tested in accordance with UL 723. Liner adhesive as recommended by manufacturer and complying with NFPA 90A and NFPA 90B, clip pin also allowable.

b. Thermal conductivity no greater than 0.035 Watt per meter-kelvin at 24 degree C.

c. Acceptable Manufacturers:

1) AEROFLEX Breathe EZ.

2) Equal performance & quality approved make

3. Acoustical Performance:

a. Determine acoustical performance with a Type F25 mounting in accordance with ASTM C423 Standard Text Method for Sound Absorption and Sound Absorption Coefficients by the Reverberation Room Method.

b. Provide minimum acoustical performance as follows:

SOUND ABSORPTION COEFFICIENT AT FREQUENCY								
Internal Duct Liner Type	**Thickness (mm)**	**125**	**250**	**500**	**1,000**	**2,000**	**4,000**	**NRC**
L1	25	0.05	0.25	0.57	0.78	0.87	0.89	0.60
L1	38	0.20	0.46	0.82	0.93	0.95	0.91	0.80
L1	50	0.20	0.57	0.90	1.02	0.96	0.92	0.90
L2	25	0.04	0.24	0.56	0.83	0.92	0.97	0.65
L2	38	0.10	0.41	0.89	0.95	0.92	0.93	0.80
L2	50	0.17	0.67	1.05	1.02	0.95	0.96	0.95

2.4 FACTORY APPLIED JACKETS

A. Insulation system datasheet indicates factory-applied jackets on various applications. When factory applied jackets are indicated, comply with the following:

1. ASJ: White, Kraft-paper, fiberglass reinforced scrim with aluminum foil backing, complying with ASTM C1136, Type I.
2. ASJ-SSL: ASJ with self-sealing, pressure sensitive, acrylic based adhesive covered by a removable protective strip; complying with ASTM C1136, Type I.
3. FSK: Aluminum foil, fiberglass reinforced scrim with kraft paper backing, complying with ASTM C1136, Type II.
4. FSP: Aluminum foil, fiberglass reinforced scrim with polyethylene backing, complying with ASTM C1136, Type II.
5. PVDC-I: 4 mil thick, white PVDC biaxially oriented barrier film with a permeance at 0.02 perms when tested according to ASTM E96 and with a flame spread index of 5 and a smoke developed index of 20 when tested in accordance with ASTM E84.
6. PVDDC-O: 6 mil thick, white PVDC biaxially oriented barrier film with a permeance at 0.01 perms when tested according to ASTM E96 and with a flame

spread index of 5 and a smoke developed index of 25 when tested in accordance with ASTM E84.

7. PVDC-SSL: PVDC jacket with self-sealing, pressure sensitive, acrylic based adhesive covered by a removable protective strip.

2.5 FIELD APPLIED JACKETS

A. Provide field-applied jackets complying with ASTM C921, Type I, unless otherwise indicated.

B. FSK Jacket: Aluminum-foil-face, fiberglass-reinforced scrim with kraft-paper backing.

C. PVC Jacketing and Pre-Formed Fitting Covers: High-impact-resistant, UV-resistant PVC complying with ASTM D1784, Class 16354-C; roll stock ready jacketing for shop or field cutting and forming, and pre-formed fitting covers. Minimum thicknesses must be 15 mils.
   1. Products: Subject to compliance with requirements, provide one of the following:
      a. Johns Manville; Zeston, 300 Series if outdoors, 2000 series if indoors.
      b. P.I.C. Plastics, Inc.; FG Series.
      c. Proto PVC Corporation; LoSmoke.
      d. Speedline Corporation; SmokeSafe.
   2. Adhesive: As recommended by jacket material manufacturer.
   3. Color: White.
   4. Factory-fabricated fitting covers to match jacket if available; otherwise, field fabricate.
      a. Shapes: 45- and 90-degree, short- and long-radius elbows, tees, valves, flanges, unions, reducers, end caps, soil-pipe hubs, traps, mechanical joints, and P-trap and supply covers for lavatories.

D. AL, Aluminum Jacket: Comply with ASTM B209, Alloy 3003, 3005, 3105 or 5005, Temper H-14.
   1. Products: Subject to compliance with requirements, provide one of the following:
      a. Childers Products, Division of ITW; Metal Jacketing Systems.
      b. PABCO Metals Corporation; Surefit.
      c. RPR Products, Inc.; Insul-Mate.
   2. Ducts and Plenums, Exposed, up to 900 mm in Diameter or with Flat Surfaces up to 1200 mm:
      a. Aluminum, Smooth: 0.5 mm thick (no corrugations.)
   3. Ducts and Plenums, Exposed, up to 1200 mm in Diameter or with Flat Surfaces up to1800 mm:
      a. Aluminum, Smooth: 0.6 mm thick (no corrugations.)
   4. Ducts and Plenums, Exposed, Larger Than 1200 mm in Diameter or with Flat Surfaces Larger Than 1800 mm:
      a. Aluminum, Smooth with 31.75 mm x 6.35 mm deep corrugations: 0.8 mm inch thick.
   5. Moisture Barrier for all Applications: 3-mil thick, heat-laminated polyethylene and DuPont’s Surlyn® coating.
   6. Field fabricated fitting covers only if factory-fabricated fitting covers are not available.

E. SS, Stainless Steel Jacket: Comply with ASTM A167 or ASTM A240/A240M.
   1. Sheet and roll stock ready for shop or field sizing.
   2. Ducts and Plenums, Exposed, up to 1200 mm in Diameter or with Flat Surfaces up to 1800 mm:
      a. Stainless Steel, Type 304 or 316, Smooth 2B Finish: 0.5mm thick.
   3. Ducts and Plenums, Exposed, Larger Than 1200 mm in Diameter or with Flat Surfaces Larger Than 1800 mm:
      a. Stainless Steel, Type 304 or 316, Smooth, with 31.75 mm x 6.35 mm deep corrugations: 0.5 mm thick.
   4. Moisture Barrier for all Applications: 3-mil thick, heat-laminated polyethylene and DuPont's Surlyn® coating.
   5. Factory-Fabricated Fitting Covers:
      a. Same material, finish, and thickness as jacket.
      b. Preformed 2-piece or gore, 45- and 90-degree, short- and long-radius elbows.
      c. Tee covers.
      d. Flange and union cover.
      e. End caps.
      f. Beveled collars.
      g. Field fabricated fitting covers only if factory-fabricated fitting covers are not available.

2.6 REMOVABLE INSULATION COVERS

A. Acceptable Manufacturers:
   1. Advance Thermal Corp.
   2. Thermal Energy Products, Inc.
   3. Temptec.
   4. Remco Technology, Inc.

B. Removable ceramic blanket type with Velcro tabs and box-stitched, 38 mm wide, D-ring straps, gussets, hot face inner jacketing, type 304 stainless steel tag with laser engraved data riveted to body, outer jacketing, type 304 stainless steel quilting pins, specifically shaped and constructed for insulated item.

2.7 ACCESSORIES

A. Adhesives:
   1. Provide materials compatible with insulation materials, jackets, and substrates and for bonding insulation to itself and to surfaces to be insulated, unless otherwise indicated.
   2. Adhesives to be waterproof, fire-retardant type.
   3. For indoor applications, use food grade adhesive for Flexible Elastomeric, ASJ, and PVC Jacket that has a VOC content of 250 g/L or less and for Mineral-Fiber Adhesive that has a VOC content of 250 g/L or less when calculated according to 40 CFR 59, Subpart D (EPA Method 24).

B. Lagging Adhesive:

1. Provide materials compatible with insulation materials, jackets, and substrates and for bonding insulation to itself and to surfaces to be insulated, unless otherwise indicated.
2. VOC content not to exceed 250 g/L.
3. Description: Comply with MIL-A-3316C Class I, Grade A and compatible with insulation materials, jackets, and substrates.
4. Provide nonflammable and fire-resistant lagging adhesives that have a maximum flame spread index of 25 and a maximum smoke developed index of 50 when tested in accordance with ASTM E84. Ensure adhesive is pigmented white and suitable for bonding fibrous glass cloth to faced and unfaced fibrous glass insulation board; for bonding cotton brattice cloth to faced and unfaced fibrous glass insulation board; for sealing edges of and bonding glass tape to joints of fibrous glass board; for bonding lagging cloth to thermal insulation; or Class 2 for attaching fibrous glass insulation to metal surfaces. Apply lagging adhesives in strict accordance with the manufacturer's recommendations for duct insulation.

C. Joint Sealants:
1. Provide materials compatible with insulation materials, jackets, and substrates.
2. Permanently flexible, elastomeric sealant.
3. Service Temperature Range: Minus 37.7 to plus 93.3 deg C.
4. For indoor applications, use food grade sealants that have a VOC content of 250 g/L or less when calculated according to 40 CFR 59, Subpart D (EPA Method 24).

D. FSK and Metal Jacket Flashing Sealants:
1. Products: Subject to compliance with requirements, provide one of the following:
   a. Childers Products, Division of ITW; CP-76-8.
   b. Foster Products Corporation, H. B. Fuller Company; 95-44.
   c. Vimasco Corporation; 750.
2. Provide materials compatible with insulation materials, jackets, and substrates.
3. Fire and water-resistant, flexible, elastomeric sealant.
4. Service Temperature Range: Minus 4.4 to plus 121.1 deg C.
5. Color: Aluminum.
6. For indoor applications, use sealants that have a VOC content of 250 g/L or less when calculated according to 40 CFR 59, Subpart D (EPA Method 24).

E. ASJ Flashing Sealants, and Vinyl, PVDC, and PVC Jacket Flashing Sealants:
1. Provide materials compatible with insulation materials, jackets, and substrates.
2. Fire and water-resistant, flexible, elastomeric sealant.
3. Service Temperature Range: Minus 4.4 to plus 121.1 deg C.
4. Color: White.
5. For indoor applications, use sealants that have a VOC content of 250 g/L or less when calculated according to 40 CFR 59, Subpart D (EPA Method 24).

F. Mastics:
1. Provide materials compatible with insulation materials, jackets, and substrates; comply with MIL-C-19565C.
2. VOC content not to exceed 100 g/L.

G. Welded Fasteners: Galvanized steel pin and washer fastener.

H. Tie Wires: Galvanized, 20 gauge.

I. Joint Tape: Foil scrim kraft (FSK) tape or (ASJ) tape to match ductwork jacketing.

J. Impale Anchors: Galvanized steel, 13-gauge nail, self-adhesive pad, 30-gauge galvanized steel washers.

PART 3 EXECUTION

3.1 FIELD PREPARATION

A. Install external insulation materials after ductwork has been cleaned, sealed, and has passed pressurization tests.

B. Clean surfaces to encourage adherence of adhesives.

3.2 INSTALLATION

A. General: Install materials in accordance with the manufacturer’s instructions.

B. External Duct Insulation for Indoor Applications:
   1. Application:
      a. Secure insulation to ductwork using wires and adhesive. Type B insulation may also be secured with mechanical fasteners, such as impale anchors.
      b. Install without sag on the underside of ductwork using adhesive or mechanical fasteners where necessary. Bend over nail tips after installation of washers to facilitate sealing of vapor barrier and to preclude injury. Stop and point insulation around access doors and damper operators to allow operation without disturbing wrapping.
      c. Lap joints and seams of external insulation a minimum of 3 inches overlaps and staple 3 inches on center. Wire where required. Utilize adhesive. Where insulation types meet, lap external insulation a minimum of 6 inches overlap and seal the vapor barrier jacket.
      d. Seal insulation with vapor barrier adhesive or tape matching jacket type to maintain the vapor barrier. Pay attention to penetrations by mechanical fasteners, joints and seams and to ends and edges.
   2. 2-Inch-Thick Type A Insulation: Band insulation with wire at 12 inches on center to reduce thickness to 1 inch at the bands.
   3. Lined Ductwork: External insulation is not required where ductwork or plenums are internally lined or constructed of fiberglass duct board, except where otherwise noted.

C. External Duct Insulation, Outdoor:
   1. Application:
      a. Install insulation as described for Types B and C, indoor.

b. Install aluminum outer jacket to enclose the duct insulation.

2. Jacket Installation:
   a. Provide seamless top and cover duct tops and sides with continuous sheet if possible.
   b. Cross break top sheet or provide slope to allow water drainage. Low points on ducts that allow standing water are not acceptable.
      1) Lap end seams 50 mm minimum and overlap in the direction of water flow to shed water.
      2) Form 20 mm-high standing seam closure at edges; install seam closure strip with mastic and rivet at 300 mm on center.
      3) Apply mastic at overlapped end seams and install gasketed screws at 150 mm on center.
      4) Extend side jacket panels to allow 25 mm rain drip lip at bottom duct closure jacket, seal with mastic, and install gasketed screws at 150 mm on center along bottom of side panel.
      5) Maintain complete integrity of closure system to shed rainwater.

D. Internal Duct Liner (Types L1 and L2):
   1. General: Apply liner in accordance with the SMACNA Duct Liner Application Standard.
   2. Application: Apply internal insulation to flat sheet metal with continuous coverage of adhesive. Use adhesive on butted edges. Install clip-pins at 380 mm on center and no more than 50 mm maximum from cut or exposed edge. Select pin length based on the liner manufacturer’s recommendations. Use of nail-type fasteners is prohibited.
   3. Dimensions: Duct dimensions shown are net inside dimension. Increase the size of sheet metal to allow for liner thickness.
   4. Plenums: Provide 25 mm mesh hardware cloth facing over attenuated plenum floors.
   5. Nosing Strips: Provide nosing strips at each joint in the duct liner where air velocities exceed 7.62 m/s, at cut edges in lined plenums, at leading edges on air intakes, and where noted or otherwise specified.

E. Existing Systems:
   1. Repair existing insulation damaged during installation of work in existing areas.
   2. Make neat connections where new and existing insulation meet.
   3. Where existing piping, ductwork, or equipment is removed, repair existing insulation surfaces neatly to match existing.

3.3 SUBMITTAL SCHEDULE

ITEM NO.	**SUBMITTAL REQUIREMENT**	**WITH BID**	**AS INDICATED**

ITEM NO.	SUBMITTAL REQUIREMENT	WITH BID	AS INDICATED
23 07 13-01	Product datasheets for insulation, liners, and adhesives.	X	
23 07 13-02	Shop drawings of duct, indicating location and types of insulation to be used.		Prior to delivery
23 07 13-03	Manufacturer’s instructions for installation.		Per construction schedule
23 07 13-04	Coordinate submittal requirements with Sustainable Design Requirements, refer to Section 40 05 01 – Basic Mechanical Requirements.		Per construction schedule.

END OF SECTION